

Anvesh Varma Dantuluri

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Education

New York University

New York, USA

Master of Science, Computer Engineering

Aug 2024 – May 2026

- **Key Coursework:** Machine Learning, Deep Learning, Reinforcement Learning, Big Data, Advanced Python for Data Science, Computer System Architecture, Applied Matrix Theory, Systems Engineering
- Migrated legacy Python CGI scripts to a modern **Django web application**, deployed using **Apache + mod_wsgi**, improving maintainability and runtime efficiency
- **Certifications:** Data Analysis with Python (IBM), Python Data Structures (University of Michigan), Google Data Analytics (Google), Data Visualization with Tableau (UC Davis)

Technical Skills

Programming & Data: Python, SQL, C++, Bash

Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Neural Networks, CNNs, Transfer Learning

LLMs & Generative AI: Large Language Models, Prompt Engineering, RAG, Embeddings

MLOps & Deployment: Model Versioning, Experiment Tracking, Docker, CI/CD, Inference Optimization

Cloud & AI Systems: AWS (EC2, S3, Lambda, Bedrock), GPU Computing, Cost Optimization

Work Experience

INFOLOB

Remote

Software Development Engineer Intern (AI Systems)

May 2025 – Aug 2025

- Designed a fault-tolerant AI orchestration layer on MCP servers integrating AWS Bedrock for inference
- Implemented autonomous fallback and recovery mechanisms increasing task completion by **33%** and availability by **40%**
- Optimized end-to-end inference pipelines sustaining sub-minute latency under high-concurrency workloads
- Reduced operational cost by **28%** through intelligent model routing and dynamic endpoint selection

INFOTECH LABS

Hyderabad, India

Software Engineering Intern (Data & Analytics)

May 2024 – Jul 2024

- Developed scalable Power BI dashboards using SQL and DAX accelerating insight delivery by **20%**
- Analyzed high-volume sales datasets identifying trends driving a measurable **15%** KPI improvement
- Automated reporting pipelines by translating ambiguous requirements into resilient data workflows
- Improved data accuracy by **30%** through validation logic and pipeline reliability enhancements

P99SOFT

Hyderabad, India

Software Development Engineer Intern

May 2023 – Jul 2023

- Delivered a production-grade Django–MySQL platform reducing operational processing time by **25%**
- Engineered asynchronous backend services enabling efficient ingestion of growing data volumes
- Scaled system capacity supporting a **50%** traffic increase without response degradation
- Collaborated across 10+ stakeholder teams to ship reliable features aligned with business priorities

Projects

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Python, Flask

- Designed a Flask-based web application for resume and cover letter generation with LaTeX PDF output
- Optimized ATS keyword matching workflows, reducing resume customization effort by **90%**
- Enabled offline language model inference to achieve **95%+ keyword match accuracy** with low latency

AI Deep Learning Library

Python, NumPy

- Built a deep learning framework from scratch with reverse-mode autodiff and a PyTorch-like API
- Engineered tensors and computational graphs enabling vectorized forward and backward operations
- Trained transformer and GAN models validating gradients via numerical checks and visualization

Adversarial Robustness Analysis

PyTorch, ImageNet

- Implemented FGSM, PGD, and patch-based attacks reducing ResNet-34 Top-1 accuracy to **0.00%**
- Evaluated cross-model transferability where patch attacks dropped DenseNet-121 accuracy by **44.8%**
- Tuned attack parameters and analyzed Top-1/Top-5 degradation trends to assess model robustness limits